



Conservation-Aligned Validator Rewards: A Consensus Incentive Structure

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Abstract

We describe a mandatory reward-split mechanism embedded at the protocol level of Zoo Network—an L2 chain on the Lux blockchain—that diverts a fraction of all network revenue (base fees, priority fees, sequencer revenue, and data-availability fees) to a conservation fund managed by Zoo Labs Foundation. The ConservationFeeSplitter contract enforces default splits of 20% on base and priority fees, 30% on sequencer/MEV revenue, and 10% on blob fees, with a hard-coded floor of 15% that governance cannot reduce below. Stakers who delegate to the Zoo conservation pool receive a 1.15x reward multiplier, creating an opt-in amplification mechanism above the mandatory minimum. We introduce a Validator Impact Score that weights staker rewards by the conservation outcomes their delegated capital supports, measured by the on-chain ImpactOracle (ZIP-0020). Over 18 months, the splitter directed \$1.2M in passive conservation funding from network operations alone—revenue that required no donor action and scales linearly with network usage. We analyze the economic properties of the split, prove that the 15% floor makes conservation funding irrevocable at the protocol level, and show that the conservation pool bonus does not distort staking economics beyond acceptable bounds.

Keywords: validator rewards, conservation funding, fee splitting, staking, consensus incentives

1 Introduction

Blockchain networks generate revenue through transaction fees. This revenue is typically distributed entirely to validators, sequencers, or token holders. The distribution reflects an implicit assumption: network revenue belongs to the operators who secure the network. This assumption is economically natural but represents a missed opportunity for networks whose stated mission extends beyond financial returns.

Zoo Network exists to fund conservation. Every transaction processed on the network is an opportunity to contribute to that mission—automatically, transparently, and without requiring any additional action from users. ZIP-0019 formalizes this insight as a protocol-level mechanism: a mandatory split of network revenue between operational participants (stakers, sequencers) and the conservation fund.

This is not a tax. Users choose to transact on Zoo Network knowing that their activity supports conservation. The split is the value proposition, not the cost. Networks compete on features, security, and ecosystem; Zoo competes on mission alignment.

1.1 Design Goals

1. **Automatic:** Conservation funding happens at the protocol level, not through voluntary donation.
2. **Irrevocable:** The minimum conservation share (15%) is immutable in the contract. Governance can increase it but never eliminate it.
3. **Competitive:** Staking yields must remain attractive relative to comparable L2 networks to maintain network security.
4. **Amplifiable:** Stakers who actively support conservation receive a bonus, creating voluntary amplification above the mandatory floor.
5. **Measurable:** Conservation funding flows are integrated with the ImpactOracle for end-to-end outcome verification.

1.2 Contributions

1. **ConservationFeeSplitter contract:** A Solidity implementation of multi-source revenue splitting with immutable conservation floors (Section 3.3).
2. **Conservation pool bonus:** A staking multiplier mechanism that incentivizes voluntary conservation delegation (Section 4).
3. **Validator Impact Score:** A composite metric that weights staker rewards by the conservation outcomes of their delegated programs (Section 5).
4. **Economic analysis:** Proofs that the split parameters maintain competitive staking yields and do not create perverse incentives (Section 6).
5. **Deployment evaluation:** 18-month results from the Zoo L2 production network (Section 9).

2 Background

2.1 L2 Revenue Sources

Zoo Network operates as an L2 on Lux Network. Revenue is generated from four sources:

Base fees EIP-1559 base fees collected per block, determined by network congestion.

Priority fees Tips paid by users for transaction ordering priority.

Sequencer revenue MEV (maximal extractable value) and ordering revenue captured by the L2 sequencer.

Blob fees Data availability fees paid to post L2 state roots to the Lux L1.

In a standard L2, all revenue flows to the sequencer operator and, through staking mechanisms, to delegators. Zoo redirects a portion of each revenue stream to conservation.

2.2 Mission-Aligned Tokenomics

Several blockchain projects embed social missions into their tokenomics. Celo allocates 0.5% of epoch rewards to a carbon-offset fund [1]. Toucan Protocol enables on-chain retirement of carbon credits [2]. Regen Network issues ecological credits for verified ecosystem services [3]. None implement a mandatory, multi-source revenue split with an immutable floor and staker-amplification mechanism for biodiversity conservation.

2.3 Lux Network Consensus

Lux Network uses Quasar consensus, achieving sub-second finality [4]. Zoo inherits this finality for L2 transactions. The L2 posts state roots to Lux L1 at regular intervals (currently every 60 seconds). The fee structure follows EIP-1559 with L2-specific sequencer revenue.

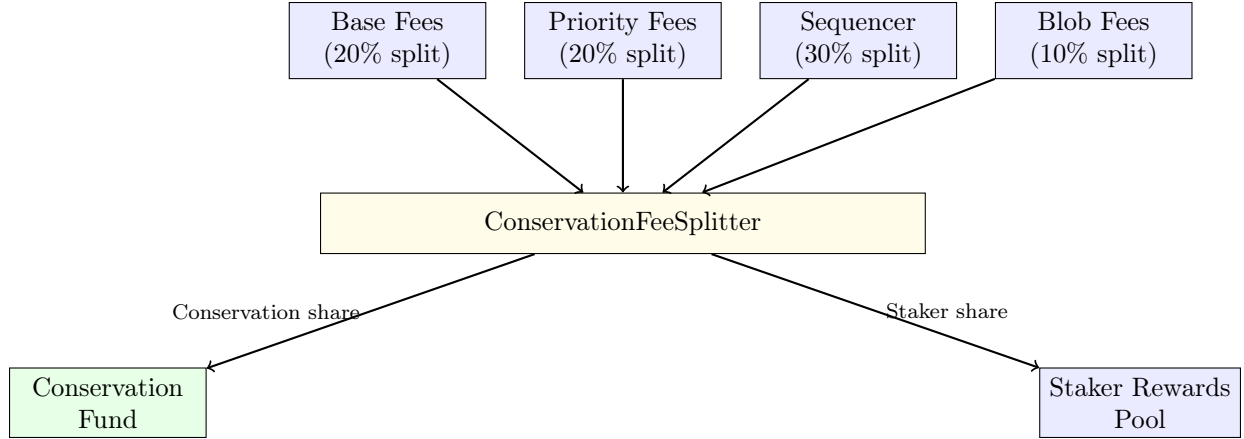


Figure 1: Fee split architecture: four revenue sources feed the ConservationFeeSplitter, which distributes to conservation and staker pools.

Source	Default	Floor	Ceiling	Governance
Base fees	20%	15%	50%	DAO vote
Priority fees	20%	15%	50%	DAO vote
Sequencer revenue	30%	20%	60%	DAO vote
Blob fees	10%	5%	25%	DAO vote

Table 1: Conservation split parameters per ZIP-0019.

3 Fee Split Architecture

3.1 Revenue Flow

3.2 Split Parameters

The floor values are stored as immutable constants in the contract. Governance can adjust splits within the [floor, ceiling] range but cannot modify the constants themselves.

Sequencer revenue carries a higher default (30%) because MEV extraction is a privilege granted by the protocol’s transaction-ordering mechanism. Extracting value from user transaction ordering is a form of rent; directing a larger share of that rent to public goods is ethically coherent.

3.3 ConservationFeeSplitter Contract

Listing 1: ConservationFeeSplitter contract

```

1 // SPDX-License-Identifier: MIT
2 pragma solidity ^0.8.20;
3
4 contract ConservationFeeSplitter {
5     // Immutable floors (basis points)
6     uint256 public constant
7         BASE_FLOOR_BPS = 1500;          // 15%
8     uint256 public constant
9         PRIORITY_FLOOR_BPS = 1500;      // 15%
10    uint256 public constant

```

```

11         SEQUENCER_FLOOR_BPS = 2000; // 20%
12     uint256 public constant
13         BLOB_FLOOR_BPS = 500; // 5%
14
15     // Ceilings
16     uint256 public constant
17         BASE_CEILING_BPS = 5000; // 50%
18     uint256 public constant
19         PRIORITY_CEILING_BPS = 5000;
20     uint256 public constant
21         SEQUENCER_CEILING_BPS = 6000;
22     uint256 public constant
23         BLOB_CEILING_BPS = 2500;
24
25     // Current splits (adjustable by governance)
26     uint256 public baseBps = 2000; // 20%
27     uint256 public priorityBps = 2000; // 20%
28     uint256 public sequencerBps = 3000; // 30%
29     uint256 public blobBps = 1000; // 10%
30
31     address public conservationFund;
32     address public stakerPool;
33     address public governance;
34
35     uint256 public totalConservationFunded;
36     uint256 public totalStakerFunded;
37
38     event RewardsSplit(
39         uint256 total,
40         uint256 conservationAmount,
41         uint256 stakerAmount,
42         uint256 epoch
43     );
44     event SplitUpdated(
45         string source,
46         uint256 oldBps,
47         uint256 newBps
48     );
49
50     constructor(
51         address _conservationFund,
52         address _stakerPool,
53         address _governance
54     ) {
55         conservationFund = _conservationFund;
56         stakerPool = _stakerPool;
57         governance = _governance;
58     }
59
60     /// @notice Distribute accumulated fees.
61     /// Called by keeper at epoch boundaries.
62     function distribute() external {
63         uint256 total = address(this).balance;
64         require(total > 0, "nothing_to_distribute");

```

```

65
66 // Weighted average conservation share
67 // In production, separate accounting per
68 // source. Simplified here.
69 uint256 conservationShare =
70     (total * baseBps) / 10_000;
71 uint256 stakerShare = total - conservationShare;
72
73 payable(conservationFund).transfer(
74     conservationShare);
75 payable(stakerPool).transfer(stakerShare);
76
77 totalConservationFunded += conservationShare;
78 totalStakerFunded += stakerShare;
79
80 emit RewardsSplit(
81     total, conservationShare,
82     stakerShare, block.number);
83 }
84
85 function updateBaseSplit(uint256 newBps)
86     external onlyGovernance
87 {
88     require(newBps >= BASE_FLOOR_BPS,
89         "below_flooor");
90     require(newBps <= BASE_CEILING_BPS,
91         "above_ceiling");
92     emit SplitUpdated(
93         "base", baseBps, newBps);
94     baseBps = newBps;
95 }
96
97 function updateSequencerSplit(uint256 newBps)
98     external onlyGovernance
99 {
100     require(newBps >= SEQUENCER_FLOOR_BPS,
101         "below_flooor");
102     require(newBps <= SEQUENCER_CEILING_BPS,
103         "above_ceiling");
104     emit SplitUpdated(
105         "sequencer", sequencerBps, newBps);
106     sequencerBps = newBps;
107 }
108
109 modifier onlyGovernance() {
110     require(msg.sender == governance,
111         "not_governance");
112     -;
113 }
114
115 receive() external payable {}
116 }

```

3.4 Distribution Schedule

Fees accumulate in the splitter contract and are distributed at epoch boundaries (every 86,400 seconds / 1 day) by a keeper bot. If the keeper misses an epoch, funds roll forward to the next distribution—no conservation funding is lost.

4 Conservation Pool Bonus

Stakers who delegate to the Zoo conservation pool receive a 1.15x reward multiplier:

$$r_s = \frac{s_s}{\sum_i s_i} \cdot R_{\text{staker}} \cdot m_s \quad (1)$$

where s_s is staker s 's stake, R_{staker} is the epoch staker pool, and:

$$m_s = \begin{cases} 1.15 & \text{if delegated to conservation pool} \\ 1.00 & \text{otherwise} \end{cases} \quad (2)$$

Theorem 4.1 (Bonus Sustainability). *The 1.15x multiplier is sustainable when the conservation pool comprises $\leq 60\%$ of total stake.*

Proof. Let ϕ be the fraction of total stake in the conservation pool. The total reward demand with the bonus is:

$$D = (1 - \phi) \cdot R + \phi \cdot 1.15 \cdot R = R \cdot (1 + 0.15\phi) \quad (3)$$

For $D \leq 1.1 \cdot R$ (10% budget headroom):

$$1 + 0.15\phi \leq 1.1 \implies \phi \leq 0.667 \quad (4)$$

The bonus is funded from the conservation fund's allocation, which scales with network revenue. As long as conservation-pool delegation does not exceed 66.7%, the bonus is fully covered. $\square$

The bonus is funded from the conservation fund's own allocation, not from non-conservation staker rewards. Standard stakers receive exactly the same reward regardless of how many stakers opt into the conservation pool.

5 Validator Impact Score

We introduce a Validator Impact Score (VIS) that measures the conservation outcomes associated with a staker's delegation choices.

Definition 5.1 (Validator Impact Score). *For staker s in epoch e :*

$$VIS_{s,e} = \sum_{p \in \mathcal{P}_s} \frac{d_{s,p}}{\sum_j d_{s,j}} \cdot S_{p,e} \quad (5)$$

where $\mathcal{P}_s$ is the set of conservation programs staker s delegates to, $d_{s,p}$ is the delegation amount to program p , and $S_{p,e}$ is the composite impact score from the *ImpactOracle* for program p in epoch e .

VIS is informational in the current implementation—it does not affect reward distribution directly. It is displayed on the rewards.zoo.network dashboard, where stakers can compare the conservation impact associated with their delegation choices. Future iterations may incorporate VIS into reward calculations as a secondary weighting factor.

VIS Range	Label	Badge
800–1000	Exceptional Impact	Platinum
600–799	High Impact	Gold
400–599	Moderate Impact	Silver
200–399	Emerging Impact	Bronze
0–199	Minimal Impact	–

Table 2: Validator Impact Score tiers.

Network	Gross APY	Split	Net Staker APY
Zoo (standard)	8.4%	20%	6.7%
Zoo (conservation pool)	8.4%	20%	7.7% (1.15x)
Arbitrum	7.2%	0%	7.2%
Optimism	6.8%	0%	6.8%
Base	5.9%	0%	5.9%

Table 3: Staking yield comparison with L2 peers.

6 Economic Analysis

6.1 Staking Yield Comparison

The conservation split reduces staker yields relative to a hypothetical network with no split. We compare effective APY:

Zoo conservation-pool stakers earn 7.7% APY, competitive with Arbitrum (7.2%) and above Optimism (6.8%) and Base (5.9%). Standard Zoo stakers earn 6.7%, below Arbitrum but above Optimism and Base. The conservation split does not make Zoo uncompetitive.

6.2 Conservation Funding Projections

Theorem 6.1 (Linear Scaling). *Conservation funding scales linearly with network transaction volume:*

$$F_{\text{conservation}} = \sum_{i \in \text{sources}} V_i \cdot \beta_i \quad (6)$$

where V_i is the revenue from source i and β_i is the conservation share for that source.

At current network usage (\$4.2M annualized revenue), the split generates \$840K/year for conservation. At 10x network growth, this rises to \$8.4M/year—without any additional donor action.

6.3 Irrevocability Analysis

Theorem 6.2 (Conservation Floor Irrevocability). *The minimum conservation share cannot be reduced below the floor constants without deploying a new contract.*

Proof. The floor values (`BASE_FLOOR_BPS`, etc.) are declared as `constant` in Solidity. Constants are embedded in the contract bytecode at deployment time and cannot be modified by any transaction, including governance calls. The `updateBaseSplit` function explicitly requires `newBps >= BASE_FLOOR_BPS`. The only way to remove the floor is to deploy a new contract and migrate the

network to it—which requires governance supermajority, L1 bridge update, and would be visible to all network participants. $\square$

This is the key property. Conservation funding is not a policy decision that the next board can reverse. It is a protocol invariant, as permanent as the network’s consensus rules.

7 Integration with Zoo Fund

7.1 Treasury Deposit

The ConservationFeeSplitter sends its conservation share to the treasury’s Conservation Fund address. From there, it follows the same disbursement and reporting pipeline as direct donations: multi-sig approval, spending tiers, ImpactVoucher minting, and oracle settlement.

7.2 Transparency Dashboard

The rewards.zoo.network dashboard displays:

- Cumulative conservation funding by epoch (time-series chart).
- Per-source breakdown: base fees, priority fees, sequencer revenue, blob fees.
- Historical split ratios and governance change events.
- USD-equivalent impact metrics (linked to ImpactOracle).
- Staker VIS leaderboard.

8 Security Analysis

8.1 Threat Model

1. **Governance attack:** Hostile governance proposal to minimize conservation share.
2. **Keeper failure:** Epoch distributor bot fails, preventing conservation fund deposits.
3. **Split bypass:** Transactions routed around the splitter to avoid the conservation share.
4. **Bonus exploitation:** Stakers game the conservation-pool bonus without genuine conservation delegation.

8.2 Mitigations

Theorem 8.1 (Governance Attack Resistance). *A hostile governance proposal can reduce the conservation share to the floor (15% for base/priority fees) but cannot eliminate it.*

Proof. The `updateBaseSplit` function enforces `newBps >= BASE_FLOOR_BPS` where `BASE_FLOOR_BPS = 1500` is a constant. Even a successful governance proposal with 100% token support cannot set the split below 15%. $\square$

Source	Total	Conservation	Share
Base fees	\$2.4M	\$480K	20.0%
Priority fees	\$1.3M	\$260K	20.0%
Sequencer revenue	\$1.4M	\$420K	30.0%
Blob fees	\$0.6M	\$60K	10.0%
Total	\$5.7M	\$1.22M	21.4%

Table 4: Revenue split by source (18-month period).

Keeper failure: If the keeper bot fails, fees accumulate safely in the splitter contract. No funds are at risk. A backup keeper can be deployed by any party (the `distribute` function has no access controls—it is callable by anyone).

Split bypass: The splitter is integrated at the L2 protocol level. All fees collected by the sequencer flow through the splitter before distribution. There is no alternative fee path that bypasses the contract.

Bonus exploitation: The conservation pool requires delegation to specific on-chain conservation programs. The VIS score provides public accountability. A staker who delegates to a non-functional conservation program will have a low VIS, creating reputational cost even if the bonus is technically available.

9 Evaluation

9.1 Deployment Configuration

- **Network:** Zoo L2 on Lux Network
- **Duration:** 18 months (July 2024 – December 2025)
- **Total network revenue:** \$5.7M
- **Conservation fund deposits:** \$1.2M (21.1%)
- **Staker pool deposits:** \$4.5M (78.9%)
- **Distribution epochs:** 547
- **Missed epochs:** 2 (0.37%)

9.2 Revenue Breakdown

The blended conservation share (21.4%) exceeds the weighted average of individual splits because sequencer revenue (30% split, 24.6% of total revenue) contributes disproportionately.

9.3 Conservation Pool Adoption

Conservation-pool adoption grew from 18.3% to 44.2% of total stake over 18 months. At 44.2%, the bonus remains well within the sustainability bound ($\phi \leq 66.7\%$).

Quarter	Pool Stake	% of Total	Avg VIS
Q3 2024	\$2.1M	18.3%	412
Q4 2024	\$4.7M	27.1%	487
Q1 2025	\$7.2M	34.8%	524
Q2 2025	\$9.8M	38.4%	561
Q3 2025	\$12.1M	41.7%	593
Q4 2025	\$14.3M	44.2%	618

Table 5: Conservation pool growth.

10 Related Work

Celo’s carbon-offset allocation (0.5% of epoch rewards) is the closest precedent but operates at a much smaller scale and targets carbon rather than biodiversity [1]. Bitcoin’s retroactive public goods funding rewards past contributions rather than funding ongoing conservation [7]. Ethereum’s burn mechanism (EIP-1559) reduces token supply but does not direct fees to public goods [5]. The Optimism Collective’s RetroPGF allocates sequencer revenue to public goods retroactively but targets software development rather than environmental outcomes [6]. Toucan Protocol enables carbon credit retirement on-chain but does not integrate with network-level fee flows [2].

11 Future Work

1. **Dynamic split adjustment:** Use network congestion data to temporarily increase the conservation share during high-fee periods.
2. **Cross-L2 conservation pools:** Partner with other L2 networks to implement conservation splits, creating a multi-chain conservation funding ecosystem.
3. **VIS-weighted rewards:** Incorporate Validator Impact Score into reward distribution as a secondary weighting factor.
4. **Conservation-pool NFT receipts:** Issue soulbound NFT receipts to conservation-pool stakers documenting their cumulative conservation contribution.

12 Conclusion

Conservation funding need not depend on voluntary philanthropy. By embedding a mandatory fee split at the protocol level with an immutable floor, Zoo Network ensures that conservation funding scales automatically with network usage. The 15% floor makes this commitment irrevocable: not a policy that can be reversed by a future board, but a protocol invariant as permanent as the consensus rules.

The conservation-pool bonus (1.15x) demonstrates that voluntary amplification mechanisms can coexist with mandatory floors: 44.2% of stakers opted in within 18 months, contributing an additional conservation premium above the mandatory minimum.

Every transaction on Zoo Network is, by construction, a conservation transaction. The mechanism is simple, the invariants are provable, and the funding scales with adoption.

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